

AI-DRIVEN KNOWLEDGE ENGINEERING

From Document Vaults to Governed,
Evolving Corporate Intelligence

“Artificial intelligence becomes institutionally useful when it can transform documents into traceable claims, connect those claims to recognized entities, test them against evidence, and update knowledge only through explicit governance.”

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Abstract

This paper defines *AI-driven knowledge engineering* as the disciplined conversion of heterogeneous information into structured, evidence-linked and governed institutional knowledge. The concept is developed through an investment-banking laboratory implemented in an Obsidian vault. The starting corpus contains one thousand synthetic company notes, scenario analyses, LLM profiles, M&A recommendations, committee memoranda, red-team objections and governed decisions. Although the corpus is analytically rich, much of its meaning remains trapped in prose. Buyer and target roles are mentioned rather than registered; environmental shocks are narrated rather than connected to company exposures; and revised decisions can displace earlier conclusions without preserving a machine-readable history.

The engineering process converts that document collection into an operational knowledge system. It preserves the source vault, creates a verified recovery copy and performs all enrichment in a separate destination. It combines deterministic extraction and calculation with bounded LLM judgment, creates typed nodes for companies, shocks, recommendations, committees, decisions, events and reassessments, and validates every relationship through an edge registry. A two-round M&A example demonstrates the complete lifecycle: a credit shock changes the attractiveness of a proposed transaction; a committee turns analysis into a governed decision; a later innovation event triggers reassessment; and a new decision supersedes the earlier one without deleting institutional history.

The pedagogical objective is to show Master of Finance students that deploying AI is not equivalent to adding a chatbot to a document store. Useful AI depends on engineered representations, explicit provenance, reproducible calculations, quality controls, temporal state and human authorization. The resulting system does not merely retrieve what was written. It can explain which claims were derived, what evidence supported them, how decisions changed corporate state and which event caused the institution to reason again.

Keywords: AI-driven knowledge engineering; knowledge graph; Obsidian; institutional intelligence; investment banking; M&A; autonomous agents; decision provenance; governance; auditability.

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1 Introduction: From Stored Information to Engineered Intelligence

An investment bank continuously produces information. Company profiles describe business models and financial conditions. Valuation models estimate outcomes. Strategists articulate industrial logic. Committees compare alternatives. Risk officers record objections. Markets move, financing conditions change and prior conclusions are revised. Each activity produces another document, table or model output.

Artificial intelligence can search, summarize and draft across this material, but fluent access is not the same as reliable knowledge. A memorandum may state that Company A should acquire Company B while leaving the buyer, target, scenario, reviewing committee and institutional outcome implicit. A person can infer those roles from context. A machine may retrieve the paragraph without knowing which entities are authoritative, whether the proposal remains active, or which later decision superseded it.

AI-driven knowledge engineering addresses this gap. Its purpose is to design the pipeline through which source material becomes claims, claims become structured relations, relations become validated graph state, and graph state becomes an input to governed reasoning. The discipline therefore combines data modelling, ontology design, information extraction, entity resolution, quantitative transformation, provenance, evaluation, workflow control and human approval.

The central proposition

AI does not create institutional knowledge merely by producing plausible text. Knowledge is engineered when a claim is attached to recognized entities, supported by traceable evidence, checked against explicit rules, assigned an epistemic status and admitted through an accountable decision process.

The paper presents a concrete implementation. Starting from a mature Obsidian vault, the workflow protects source evidence, derives a graph-enriched destination and uses AI only within bounded stages. It progresses from *documents containing possible knowledge* to *validated knowledge objects that people and agents can inspect, calculate, traverse and govern*.

2 What Is AI-Driven Knowledge Engineering?

2.1 A working definition

AI-driven knowledge engineering is the deliberate design and operation of processes that:

1. acquire and preserve authoritative source material;
2. extract candidate entities, claims, events and relationships;
3. resolve references to stable institutional identifiers;
4. map extracted content into a controlled ontology;
5. calculate derived properties through visible, reproducible methods;
6. evaluate syntactic, referential, semantic and evidential quality;
7. distinguish AI-generated candidates from adopted knowledge;
8. update state through versioned decisions and explicit permissions;
9. expose knowledge to people, analytics and bounded agents; and
10. retain the audit evidence required to reconstruct every change.

The engineered knowledge state can be written compactly as

$$K_t = (V_t, E_t, \Theta_t, \Pi_t, \Gamma_t),$$

where V_t is the set of nodes at time t , E_t the set of typed relationships, Θ_t the properties and epistemic states, Π_t the provenance records, and Γ_t the policies and validation constraints. New information does not enter this state automatically. It passes through a controlled transformation:

$$K_{t+1} = \Phi(K_t, o_t, m_t, d_t, \Gamma_t),$$

where o_t is a new observation, m_t a model-generated candidate and d_t an institutional decision. The transformation Φ includes extraction, normalization, validation, review and state update.

2.2 Documents, graphs and AI-driven knowledge engineering

Table 1: Three layers of an institutional knowledge system

Layer	Primary object	Central question	Example
Document layer	File, record or model output	What information has been captured?	Company note, scenario memo, committee report
Knowledge graph	Entity, relation and property	How are recognized objects connected?	Company $\rightarrow$ proposed buyer in $\rightarrow$ recommendation
Knowledge engineering	Pipeline, model, graph, validation and governance	How does information become reliable operational knowledge?	AI extracts a claim; code validates it; a committee adopts or rejects it

The graph is therefore one engineered artifact, not the whole discipline. A graph may contain a relation called *supersedes*; knowledge engineering defines how that relation is proposed, validated, authorized, versioned and propagated to dependent objects. It also defines how the relation can be reproduced from source evidence if the graph must be rebuilt.

3 Why the Mature Vault Still Required Engineering

The maturity-ladder vault already contained a substantial corporate corpus: one thousand companies, strategic profiles, nine joint macro-innovation scenarios, recommendations, specialist analyses, committee deliberations, dissent, decisions, external-adviser opinions and audit records. The problem was not a lack of content. It was the absence of a dependable process for converting that content into shared, computable and governable state.

Four engineering gaps limited the usefulness of AI and automation.

3.1 Important claims remained embedded in prose

A memorandum might state “C0123 acquires C0456.” An LLM could summarize the sentence, yet downstream systems still needed to identify the two companies, assign buyer and target roles, link the claim to its source and determine whether it represented an idea, a recommendation or an approved transaction. Without that transformation, the relation was readable but not operational.

3.2 Narrative scenarios were not computably connected to companies

The vault described a credit seizure and an AI-productivity wave. Company notes contained leverage, credit and innovation measures. However, no common exposure model connected a scenario vector to each company’s sensitivity vector. AI could discuss the shock qualitatively, but the system could not reproduce which firms were most affected or trace that result into transaction decisions.

3.3 Analytical proposals and authorized decisions were conflated

A recommendation is an analytical proposal; a decision is an institutional act. If both appear only in a final memo, an AI system may treat persuasive language as authority. The knowledge model must distinguish who proposed an action, who reviewed it, what outcome was adopted and what conditions limit reliance.

3.4 Knowledge evolution was not represented explicitly

The latest memo could replace the practical relevance of an earlier memo without creating a supersedes relation. Retrieval might return both documents as equally current. Institutional learning then appeared as a stack of files rather than a traceable sequence of observations, judgments and state transitions.

A document can be accurate and an extracted claim can be plausible while both remain operationally orphaned. If referenced companies, scenarios, evidence and decisions are not resolved to validated objects, the system cannot determine what depends on the claim, what would change if it were corrected, or whether an agent is permitted to use it.

4 The Three-Vault Knowledge-Engineering Pipeline

The engineering workflow begins with preservation rather than model inference.

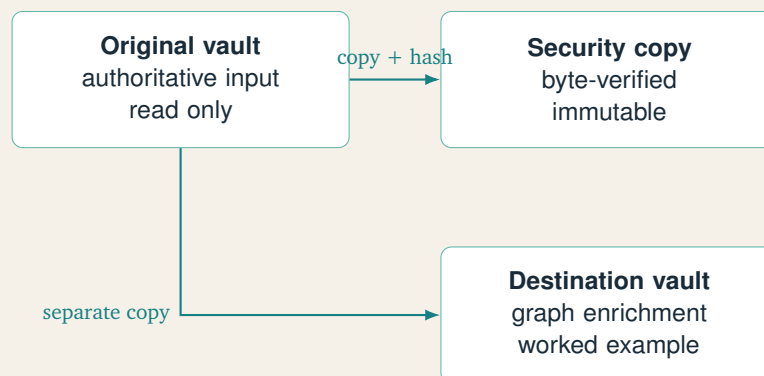


Figure 1: Preservation precedes AI-assisted knowledge transformation.

The notebook computes SHA-256 hashes for every file in the original vault. It creates a timestamped security copy and a separate timestamped destination. Both are compared with the source manifest before extraction or enrichment begins. After the engineering run, the source and security copy are hashed again. Any unexpected change causes the run to fail.

This design separates three operational roles:

- **source authority:** the original vault remains the preserved historical corpus;
- **recovery evidence:** the security copy proves that the starting state can be reconstructed byte for byte;
- **engineered workspace:** the destination receives extracted claims, derived metrics, graph objects and AI-generated candidates.

Reversibility is a quality attribute of trustworthy AI. A knowledge pipeline must be able to show what it received, what it derived and how to return to the pre-enrichment state.

5 Engineering the Corporate Decision Model

The domain model defines the objects that the AI is allowed to recognize and the relations it may propose. This controlled ontology prevents every prompt or agent from inventing a private representation. The worked system uses seven central node types.

Table 2: Core node types

Node type	Engineering role	Representative properties
Company	Resolved corporate entity in the synthetic universe	Financials, strategy, sensitivities, outlook state
Environment shock	Structured macro or innovation hypothesis	Shock vector, description, severity, source
Recommendation	Candidate transaction thesis	Buyer, target, status, round, provenance
Committee Decision	Accountable review process Authorized advance, hold or reject outcome	Mandate, participants, review scope Rationale, conditions, confidence, authority
Event	Observed or simulated change in the environment	Trigger, time, affected entities
Reassessment	Review initiated by material new information	Prior decision, new evidence, revised decision

The principal relationships include:



The model follows the subject-predicate-object logic formalized in RDF, while retaining properties needed for weights, confidence, time and provenance. The implementation uses CSV registries and GraphML for machine processing, while Obsidian wikilinks expose a readable projection to analysts. AI components must emit candidates that conform to this model; they do not redefine the model at run time.

6 Engineering Environmental Knowledge

6.1 Transforming company data into sensitivity features

The notebook converts existing company attributes into explicit sensitivities for all 1,000 companies. The shared feature space covers:

- policy rates;
- credit spreads;
- GDP growth;

- AI productivity;
- digital substitution;
- regulated transition.

These values are deterministic transformations rather than LLM estimates. Leverage and capital intensity increase rate sensitivity. Weak credit and leverage increase spread sensitivity. Growth and beta influence cyclicity. Innovation and R&D intensity affect AI-productivity exposure. Moat and innovation inform substitution risk, while ESG, moat and credit quality shape regulated-transition resilience. Each feature therefore has an inspectable lineage from source fields to derived value.

For company i , let

$$\beta_i = (\beta_{i,r}, \beta_{i,c}, \beta_{i,g}, \beta_{i,a}, \beta_{i,d}, \beta_{i,q})$$

denote sensitivities to rates, credit, growth, AI, digital substitution and regulatory transition.

6.2 Representing shocks and computing impact

A shock s is engineered as a vector z_s over the same dimensions. The first worked state represents a credit seizure:

$$z_{M3} = (1, 1, -0.8, 0, 0, 0),$$

while the innovation event emphasizes AI productivity and substitution:

$$z_{I1} = (0, 0, 0, 1, 0.45, 0).$$

The exposure impact is:

$$I_{i,s} = \beta_i^\top z_s.$$

The measure is not a calibrated valuation model and is never presented as one. It is a transparent feature transformation that demonstrates a central knowledge-engineering principle: narrative concepts must be mapped to common, testable dimensions before their effects can be compared or propagated.

From language to computable knowledge

Before engineering, “credit conditions deteriorated” was a sentence in a scenario. After engineering, the shock is a versioned node, each company has a reproducible sensitivity vector, calculated impact becomes an edge property and affected recommendations can be found through graph traversal. AI can explain the result, but deterministic features make the result reproducible.

7 A Two-Round AI-Driven M&A Knowledge Cycle

7.1 Round 0: establish trusted baseline state

Deterministic retrieval selects one buyer-target pair from the existing screen. Entity-resolution rules bind both references to registered company nodes. Credit and innovation shocks are registered as environment nodes, and all calculated exposures retain their formulas and source fields. The baseline state contains trusted entities and derived relationships but no institutional transaction decision.

7.2 Round 1: generate, validate and decide

An AI component drafts a candidate recommendation that Company A acquire Company B. The system validates entity identifiers, registers buyer and target roles and attaches the recommenda-

tion to supporting evidence. The M3 credit-seizure node stress-tests the transaction. A committee node represents the accountable review process, while a bounded decision model returns one of three permitted outcomes: ADVANCE, HOLD or REJECT. Its structured output also contains rationale, conditions, risks, evidence identifiers and confidence.

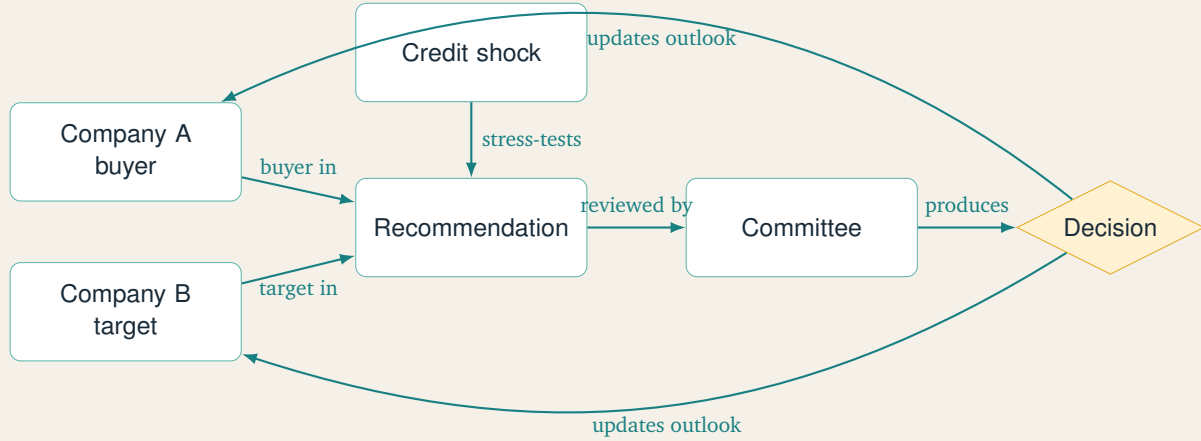


Figure 2: Round 1: AI proposes; validation and governance determine graph state.

The adopted decision becomes a first-class node rather than an unstructured sentence. It connects backward to the recommendation, evidence and committee, and forward to the outlook of both companies. The committee conclusion is not merely archived: it changes the knowledge state presented to later analysts and agents. A model-generated candidate that fails validation or authorization does not receive this status.

7.3 Round 2: detect change, reassess and supersede

A new AI-productivity event materializes from the innovation shock. The event is resolved against the controlled scenario dimensions and linked to buyer and target through separately calculated impacts. A materiality rule triggers a reassessment. The reassessment explicitly retrieves the prior decision, the new event, unresolved conditions and updated company state. A second governed decision then supersedes the first without deleting it.

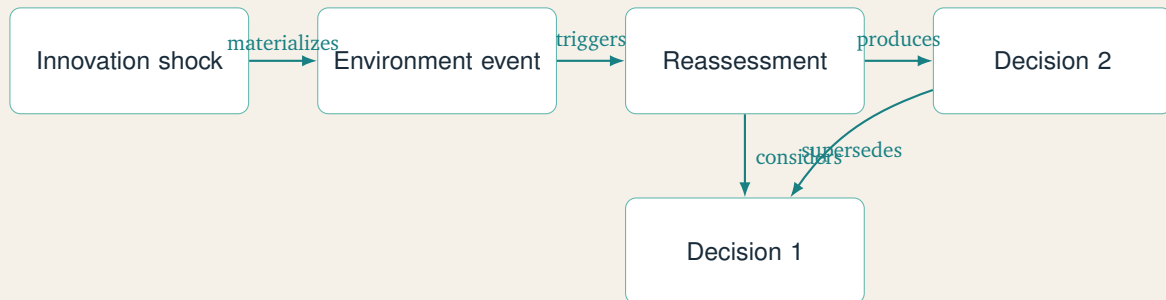


Figure 3: Round 2 turns new evidence into a governed knowledge-state transition.

This is temporal knowledge engineering. The system retains not only the latest answer but the causal sequence through which the answer changed:

recommendation → decision₁ → new event → reassessment → decision₂.

8 Decisions as Knowledge-State Operations

A decision has at least five engineering effects on institutional knowledge.

8.1 It changes state

The recommendation moves from proposed to advance, hold or reject. Buyer and target receive a new transaction-related outlook state with an effective time, source decision and authorization status.

8.2 It creates obligations

Conditions precedent become machine-visible verification requirements. A hold may require financing certainty, antitrust review or downside validation, and each requirement can later be marked open, satisfied, waived or failed.

8.3 It changes retrieval

A later agent searching for active opportunities must rank advanced, held and rejected recommendations differently. The decision therefore changes future retrieval, prompt context and the actions that the system is allowed to propose.

8.4 It preserves institutional accountability

The graph identifies which committee authorized the decision, which model or workflow contributed, and which evidence identifiers supported it. Model confidence remains a property of the output; it never substitutes for authority.

8.5 It establishes a temporal reference point

A later reassessment must state which decision it supersedes. Without that edge, an AI system may retrieve two incompatible answers without knowing which is current, why they differ or when the transition occurred.

Decision as an engineered state transition

A decision is not merely generated content attached to a graph. It is an authorized operation on institutional knowledge. It may activate, suspend, reject, condition or supersede claims and relationships, thereby changing what later agents retrieve and what people understand to be current.

9 Projecting Engineered Knowledge into Obsidian

The destination vault remains usable as ordinary Markdown. Each company note receives an additive knowledge-engineering section containing:

- its environmental sensitivity table;
- links to relevant shock nodes;
- calculated shock impacts;
- its role in the recommendation;
- links to the committee, initial decision, event, reassessment and revised decision.

The write-back is deliberately additive. Existing company facts are not silently overwritten by model output. A unique run marker, provenance block and generated status make the addition auditable. The note becomes a human-readable projection of machine-readable state rather than the sole source of that state.

The system also produces:

1. `nodes.csv`, the node registry;
2. `edges.csv`, the typed edge registry;
3. `company_environment_sensitivities.csv`;
4. `company_state_history.csv`;
5. `corporate_decision_graph.graphml`;
6. before-and-after graph images;
7. an Obsidian Decision Propagation Dashboard;
8. a graph-enrichment audit manifest.

This multi-representation design is intentional. Markdown supports analysts and retrieval-augmented models; CSV supports inspection and reproducible calculation; GraphML supports traversal and network analysis; PNG supports teaching and committee communication. The registries remain the portable basis from which every projection can be rebuilt.

10 Division of Labour Between AI and Deterministic Systems

The LLM does not freely invent institutional truth or mutate the graph. In this implementation, deterministic code selects the pair, calculates sensitivities, resolves identifiers, registers nodes, enforces allowed relations, validates endpoints, updates files and writes the audit record. The model performs a bounded synthesis and committee-style judgment over a supplied evidence packet.

The model's output contract contains:

- outcome;
- concise rationale;
- conditions precedent;
- principal risks;
- evidence identifiers;
- confidence.

Schema-constrained outputs, fixed execution limits and diagnostic fallbacks reduce operational variability. Even so, schema validity is not semantic or institutional validity. Code normalizes outcomes, checks evidence identifiers, rejects unsupported relation types and prevents an unavailable or malformed model response from authorizing any state transition.

A division of cognitive labour

The model extracts, compares and explains. Code calculates, resolves and enforces contracts. The graph stores relationships and history. Obsidian exposes the result to people. Governance determines whether generated knowledge remains a candidate or becomes institutionally adopted.

11 Quality Engineering and Orphan Control

Every generated object must satisfy explicit quality checks, and every edge must connect registered endpoints. The notebook computes:

- registered node count;
- registered edge count;
- unregistered edge endpoints;
- isolated nodes;

- confirmation that buyer and target backlinks were written.

An edge-registry row contains a stable edge ID, source, relation, target, round, provenance, weight and status. A node registry records identifiers, types, labels, source locations and epistemic status. Together they convert links from decorative navigation into testable data products.

Structural integrity does not guarantee semantic truth. A well-formed *affects* edge may still carry an inappropriate weight, and a fluent rationale may cite irrelevant evidence. AI-driven knowledge engineering therefore requires several layers of validation:

Table 3: Validation layers

Layer	Engineering question	Representative control
Syntactic	Does the output conform to its contract?	Schema validation
Referential	Do identifiers resolve to registered objects?	Entity and endpoint checks
Semantic	Is the claim meaningful in the domain model?	Ontology constraints and review
Evidential	Does the cited source support the claim?	Provenance and entailment checks
Quantitative	Can the derived value be reproduced?	Visible formula and source fields
Temporal	Are validity and supersession preserved?	Event time, effective time and version links
Governance	Was the state change authorized?	Candidate/adopted status and human gate

12 Governance of AI-Engineered Knowledge

The original corpus, security copy and destination graph serve different governance roles. AI output enters the destination as candidate knowledge. It does not become authoritative merely because it is fluent, linked, visually compelling or repeated by other generated materials.

A production knowledge-engineering system should distinguish:

1. **source evidence**, preserved as received;
2. **deterministically derived knowledge**, with reproducible formulas;
3. **AI-generated candidate knowledge**, with model and prompt provenance;
4. **reviewed and adopted institutional knowledge**; and
5. **superseded knowledge**, retained for historical reconstruction.

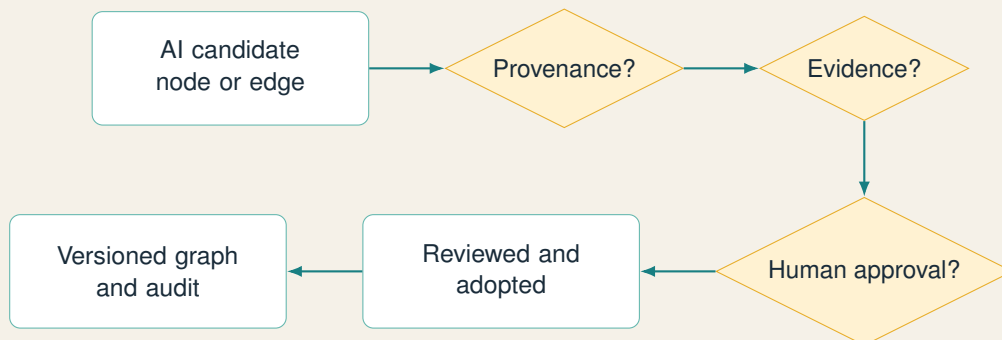


Figure 4: AI-engineered knowledge passes through epistemic and institutional gates.

The audit manifest records that the system is synthetic, recommendation-only, human-gated, non-executing and dry-run. These are not disclaimers appended after analysis. They are machine-

readable operating constraints that define what the system may generate, what people may rely on and which actions remain outside its authority.

13 AI-Driven Knowledge Engineering for Autonomous Agents

Agents require more than long context windows or vector retrieval. They require an engineered knowledge state that supports action selection, dependency analysis and permission checks. A document repository can retrieve passages; a typed, temporal and governed graph can answer questions such as:

- Which active recommendations depend on companies highly exposed to a credit seizure?
- Which decisions are eligible for reassessment after an innovation event?
- Which company outlooks were changed by a particular committee decision?
- Which current decision superseded a prior hold?
- Which conditions precedent remain unresolved?

The resulting agentic loop becomes:

observe event → traverse affected subgraph → calculate impact
→ propose reassessment → committee decision → update graph.

This is a stronger and safer form of autonomy than repeatedly prompting over documents. The agent observes validated state, traverses only relevant dependencies, proposes a bounded action, receives an institutional result and updates the state that governs its next cycle. Crucially, the agent may propose a knowledge change without possessing authority to adopt it.

14 Pedagogical Implications for Finance

For Master of Finance students, AI-driven knowledge engineering joins several disciplines that are too often taught separately.

14.1 Corporate finance

The engineered model separates strategic logic, affordability, financing resilience and environmental sensitivity, then records how each contributes to a recommendation.

14.2 Decision theory

Recommendations, evidence, decisions and state transitions become separate objects. Students can observe how new information changes an action without erasing the rationale for the earlier choice.

14.3 Risk management

The sensitivity-shock product makes heterogeneous exposure explicit. Conditions precedent, uncertainty and downside states become queryable parts of the decision record.

14.4 Institutional governance

The committee is not a narrative flourish. It is the accountable boundary at which AI-generated analysis may be accepted, conditioned, rejected or returned for further evidence.

14.5 Artificial intelligence

Students learn that an AI system is not a model alone. It is an engineered combination of models, tools, representations, evaluations, memory, contracts, permissions, observability and feedback.

Suggested seminar exercise

Give students the same buyer, target, source packet and sensitivity vectors. Ask one group to define the credit shock, another to prompt an AI recommendation, a third to validate its claims, a fourth to act as committee and a fifth to introduce a new innovation event. Require each group to identify its inputs, outputs, evidence, nodes, edges and permissions. Grade provenance, reproducibility and state management separately from the persuasiveness of the prose.

15 Failure Modes in AI-Driven Knowledge Engineering

15.1 Ontology drift

Models and agents invent new entity classes or relation names that duplicate existing meanings. Controlled vocabularies, schema versions and migration rules are required.

15.2 False precision

Generated scores and sensitivity weights may look empirical even when they are heuristic transformations. Their methodology, calibration status and permitted use must remain visible.

15.3 Circular evidence

An AI-generated note may later be retrieved as support for another AI-generated claim. Provenance must reveal when apparent corroboration is merely recursive reuse of the same unsupported assertion.

15.4 Graph amplification

A mistaken entity resolution or edge can affect many dependent recommendations. High-centrality nodes and high-impact changes require stronger validation and review thresholds.

15.5 Temporal collapse

Overwriting a claim or decision erases why institutional judgment changed. Effective dates, versioning and supersession links must preserve history.

15.6 Orphan proliferation

Extracted claims and generated nodes without validated provenance or relations accumulate as plausible-looking objects with no operational role.

15.7 Authority confusion

A machine-generated decision object may appear official. Status must distinguish simulation, draft, candidate, reviewed, adopted, rejected and superseded.

15.8 Evaluation blindness

A pipeline may report successful execution while producing poor entity matches, weak evidence alignment or unstable judgments. Engineering metrics must measure knowledge quality, not merely API availability and file creation.

15.9 Prompt and model drift

A model, prompt or retrieval configuration can change the meaning of generated outputs between runs. Versioned prompts, model identifiers, test sets and regression evaluation are required for reproducibility.

16 Design Principles for AI-Driven Knowledge Engineering

1. Preserve source evidence before extraction, inference or enrichment.
2. Separate recovery copies from experimental and production workspaces.
3. Define stable identifiers, node classes and relation types before prompting.
4. Use AI for ambiguity and synthesis; use deterministic code for calculation, validation and policy enforcement.
5. Convert material claims into typed, testable and evidence-linked objects.
6. Connect narrative scenarios and corporate data through common dimensions.
7. Distinguish recommendation, model judgment and institutional decision.
8. Treat every consequential output as a candidate until explicitly adopted.
9. Make authorized decisions update dependent state through visible rules.
10. Represent events, reassessments and supersession instead of overwriting history.
11. Retain provenance for sources, prompts, models, formulas and reviewers.
12. Validate schemas, identifiers, semantics, evidence, time and permissions.
13. Measure orphan rates, unsupported claims and downstream impact.
14. Store knowledge in portable, inspectable and rebuildable formats.
15. Keep execution authority outside analytical and generative components.
16. Design evaluation questions before scaling the knowledge graph.
17. Engineer the system for the decisions future people and agents must make.

17 Conclusion: Engineering AI into Institutional Knowledge

The journey from an Obsidian document vault to an AI-driven knowledge system begins with a change in the unit of work. The system must manage not only files, but also entities, claims, evidence, derived features, relationships, epistemic states, decisions and temporal transitions. Language models contribute valuable capabilities for extraction, comparison and explanation, but these capabilities become institutionally dependable only when embedded in an engineered process.

In the worked example, deterministic retrieval and entity resolution establish a buyer and target. An AI component produces a structured recommendation from a bounded evidence packet. A credit shock stress-tests the proposal through reproducible company sensitivities. Validation controls check identifiers, relations and evidence. A committee then produces an authorized decision that changes recommendation status and company outlook. When a later innovation event becomes material, the system triggers reassessment and records a second decision that supersedes the first without erasing it. Every stage remains visible in Markdown, CSV, GraphML, visualizations and an audit manifest.

The result is more than enhanced retrieval. Institutional memory acquires a causal, temporal and governable form. The system remembers which conclusion came first, what evidence and model contributed to it, what event changed the relevant state, who authorized reconsideration and how the revised decision altered the companies and opportunity. This is the foundation required for useful autonomous agents: not unlimited freedom to generate, but reliable access to structured state and bounded mechanisms for proposing change.

The final proposition

AI-driven knowledge engineering is the discipline of turning model capability into governed institutional intelligence. It transforms documents into evidence, language into typed claims, calculations into reproducible features, generated analysis into reviewable candidates, decisions into authorized state transitions and history into a reconstructable path of change.

A Illustrative Knowledge-Relation Registry

Source	Relation	Target	Meaning
Company A	proposed_buyer_in	Recommendation 1	Buyer role
Company B	proposed_target_in	Recommendation 1	Target role
Credit shock	stress_tests	Recommendation 1	Scenario dependency
Recommendation 1	reviewed_by	Committee 1	Governance routing
Committee 1	produced	Decision 1	Institutional outcome
Decision 1	updates_outlook	Company A	State propagation
Decision 1	updates_outlook	Company B	State propagation
Innovation shock	materializes_as	Event 2	New observation
Event 2	triggers	Reassessment 2	Reopening rule
Reassessment 2	considers_prior	Decision 1	Historical context
Reassessment 2	produces	Decision 2	Revised outcome
Decision 2	supersedes	Decision 1	Temporal ordering

B Minimum AI Knowledge-Engineering Checklist

1. Are authoritative source files preserved and content-addressed?
2. Is there a separately verified recovery copy?
3. Are extraction inputs, prompts, models and outputs versioned?
4. Are entity identifiers, node types and relation types defined?
5. Does every relation connect registered endpoints?
6. Does every material claim link to supporting evidence?
7. Are deterministic calculations reproducible from visible source fields?
8. Are company exposures linked quantitatively to environment nodes?
9. Are AI recommendations distinct from institutional decisions?
10. Do decisions identify evidence, authority, conditions and effective time?
11. Can events trigger explicit, policy-controlled reassessment?
12. Are superseded claims and decisions preserved?
13. Can every graph change be traced to source, model, code and reviewer?
14. Are candidate, adopted, rejected and superseded states distinguished?
15. Is human authorization required for consequential reliance?
16. Are quality metrics defined for extraction, resolution and evidence support?
17. Can the knowledge state be rebuilt from portable artifacts?
18. Does the audit prove that the original corpus remained unchanged?

C Compact Formal Model of the Engineering Cycle

Let C denote companies, S shocks, R recommendations, K committees, D decisions, E events and X reassessments. Let P denote provenance records and Q quality and policy constraints. The engineered knowledge state at time t is:

$$\mathcal{K}_t = (C \cup S \cup R \cup K \cup D \cup E \cup X, \mathcal{E}_t, P_t, Q_t).$$

For company i and shock s ,

$$I_{i,s} = \beta_i^\top z_s.$$

An AI model M proposes a structured recommendation from evidence packet B_t :

$$\hat{r}_t = M(B_t), \quad r_t = \text{Validate}(\hat{r}_t, Q_t).$$

Only a validated recommendation is reviewed under policy $\mathcal{P}$:

$$d_t = \delta(r_t, I_{A,s}, I_{B,s}, \mathcal{P}), \quad d_t \in \{\text{ADVANCE}, \text{HOLD}, \text{REJECT}\}.$$

If event e_{t+1} exceeds a materiality threshold, it produces reassessment x_{t+1} , and:

$$d_{t+1} = \delta(r_t, I_{A,s'}, I_{B,s'}, d_t, \mathcal{P}).$$

The knowledge-state update preserves both decisions, adds provenance and records

$$(d_{t+1}, \text{supersedes}, d_t) \in \mathcal{E}_{t+1}.$$

Thus the engineering transformation is

$$\mathcal{K}_{t+1} = \Phi(\mathcal{K}_t, e_{t+1}, M, Q_t, d_{t+1}),$$

where Φ is permitted to change adopted state only after validation and authorization.

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